

## Chapter 1 A Tour of Computer Systems

### Problem 1.1

A. We use the formula with  $\alpha = \frac{3}{5}$  and  $k = \frac{150}{100} = \frac{3}{2}$ .

$$\begin{aligned} S &= \frac{1}{\left(1 - \frac{3}{5}\right) + \frac{3}{5} \cdot \frac{2}{3}} \\ &= \frac{1}{\frac{2}{5} + \frac{2}{5}} \\ &= \frac{5}{4} \\ &= 1.25 \times \end{aligned}$$

B. We use the formula and work our way back:

$$\begin{aligned} \frac{5}{3} &= \frac{1}{\left(1 - \frac{3}{5}\right) + \frac{3}{5k}} \\ \frac{3}{5} &= \frac{2}{5} + \frac{3}{5k} \\ \frac{1}{5} &= \frac{3}{5k} \\ 1 &= \frac{3}{k} \\ k &= 3 \end{aligned}$$

So the drive through Montana needs a speedup of  $3 \times$  which is 300 km/hr.

### Problem 1.2

Use the formula with  $\alpha = \frac{4}{5}$  and  $S = 2$  and solve for  $k$ .

$$\begin{aligned} 2 &= \frac{1}{\left(1 - \frac{4}{5}\right) + \frac{4}{5k}} \\ \frac{2}{5} + \frac{8}{5k} &= 1 \\ \frac{8}{5k} &= \frac{3}{5} \\ \frac{1}{k} &= \frac{3}{8} \\ k &= \frac{8}{3} \end{aligned}$$

## Chapter 2 Representing and Manipulating Information

### Problem 2.1

- A. 0x39A7F8 to binary: 0011 1001 1010 0111 1111 1000
- B. 1100100101111011 to hexadecimal: 0xC97B
- C. 0xD5E4C to binary: 1101 0101 1110 0100 1100
- D. 1001101110011110110101 to hexadecimal: 0x26E7B5

### Problem 2.2

| $n$ | $2^n$ (decimal) | $2^n$ (hexadecimal) |
|-----|-----------------|---------------------|
| 9   | 512             | 0x200               |
| 19  | 524288          | 0x80000             |
| 14  | 16384           | 0x4000              |
| 16  | 65536           | 0x10000             |
| 17  | 131072          | 0x20000             |
| 5   | 32              | 0x20                |
| 7   | 128             | 0x80                |

### Problem 2.3

| Decimal | Binary    | Hexadecimal |
|---------|-----------|-------------|
| 0       | 0000 0000 | 0x00        |
| 167     | 1010 0111 | 0xA7        |
| 62      | 0011 1110 | 0x3E        |
| 188     | 1011 1100 | 0xBC        |
| 55      | 0011 0111 | 0x37        |
| 136     | 1000 1000 | 0x88        |
| 243     | 1111 0011 | 0xF3        |
| 82      | 0101 0010 | 0x52        |
| 172     | 1010 1100 | 0xAC        |

| Decimal | Binary    | Hexadecimal |
|---------|-----------|-------------|
| 231     | 1110 0111 | 0xE7        |

### Problem 2.4

- A.  $0x503c + 0x8 = 0x5044$
- B.  $0x503c - 0x40 = 0x4ffc$
- C.  $0x503c + 64 = 0x507c$
- D.  $0x50ea - 0x503c = 0xae$

### Problem 2.5

|    | Little endian | Big endian |
|----|---------------|------------|
| A. | 21            | 87         |
| B. | 21 43         | 87 65      |
| C. | 21 43 65      | 87 65 43   |

### Problem 2.6

A.

$0x00359141$  in binary: 0000 0000 0011 0101 1001 0001 0100 0001

$0x4A564504$  in binary: 0100 1010 0101 0110 0100 0101 0000 0100

B.

```
00000000001101011001000101000001
01001010010101100100010100000100
*****
```

There are 21 matching bits.

C.

The whole integer occurs in the float representation, except for the most-significant bit which is a 1. Similarly, some of the most-significant bits of the float representation do not occur in the int representation.

### Problem 2.7

It prints 61 62 63 64 65 66 (it does not print the terminating null character because the `strlen` function does not count it).

### Problem 2.8

| Operation | Result     |
|-----------|------------|
| a         | [01101001] |
| b         | [01010101] |
| ~a        | [10010110] |
| ~b        | [10101010] |
| a & b     | [01000001] |
| a   b     | [01111101] |
| a ^ b     | [00111100] |

### Problem 2.9

A. The following colors complement each other:

Black  $\leftrightarrow$  White

Blue  $\leftrightarrow$  Yellow

Green  $\leftrightarrow$  Magenta

Cyan  $\leftrightarrow$  Red

B.

Blue | Green = Cyan

Yellow & Cyan = Green

Red ^ Magenta = Blue

### Problem 2.10

| Step      | *x              | *y              |
|-----------|-----------------|-----------------|
| Initially | a               | b               |
| Step 1    | a               | a ^ b           |
| Step 2    | a ^ (a ^ b) = b | a ^ b           |
| Step 3    | b               | b ^ (a ^ b) = a |

### Problem 2.11

A. In the final iteration we have `first = k` and `last = k` (swap the middle element with itself).

B. In this case `*x` and `*y` point to the same address and the steps become:

| Step      | *x               | *y               |
|-----------|------------------|------------------|
| Initially | a                | a                |
| Step 1    | $a \wedge a = 0$ | $a \wedge a = 0$ |
| Step 2    | $0 \wedge 0 = 0$ | $0 \wedge 0 = 0$ |
| Step 3    | $0 \wedge 0 = 0$ | $0 \wedge 0 = 0$ |

C. We can fix it by changing the condition to `first < last` since the middle element does not need to be swapped anyway.

### Problem 2.12

- A. `x & 0xFF` leaves the least significant byte and sets everything else to zero.
- B. `x ^ ~0xFF` inverts everything except the least significant byte.
- C. `x | 0xFF` sets the least significant byte to ones and leaves everything else.

### Problem 2.13

`x | y` is equivalent to `bis(x, y)`.

`x ^ y` is equivalent to `bis(bic(x, y), bic(y, x))`.

### Problem 2.14

We have `x = 0110 0110` and `y = 0011 1001`.

| Expression              | Value                                                            | Expression                   | Value |
|-------------------------|------------------------------------------------------------------|------------------------------|-------|
| <code>x &amp; y</code>  | 0010 0000                                                        | <code>x &amp;&amp; y</code>  | 1     |
| <code>x   y</code>      | 0111 1111                                                        | <code>x    y</code>          | 1     |
| <code>~x   ~y</code>    | 1111 1111 1111 1111 1111 1111 1101 1111<br>(assuming 32-bit int) | <code>!x    !y</code>        | 0     |
| <code>x &amp; !y</code> | 0                                                                | <code>x &amp;&amp; ~y</code> | 1     |

### Problem 2.15

`!(x ^ y)` is equivalent to `x == y` because `x ^ y` will be 0 only if all the bits match.

### Problem 2.16

| $x$                 | $x \ll 3$           | $x \gg 2$ (logical) | $x \gg 2$ (arithmetic) |
|---------------------|---------------------|---------------------|------------------------|
| $0xC3 = 1100\ 0011$ | $0001\ 1000 = 0x18$ | $0011\ 0000 = 0x30$ | $1111\ 0000 = 0xF0$    |
| $0x75 = 0111\ 0101$ | $1010\ 1000 = 0xA8$ | $0001\ 1101 = 0x1D$ | $0001\ 1101 = 0x1D$    |
| $0x87 = 1000\ 0111$ | $0011\ 1000 = 0x38$ | $0010\ 0001 = 0x21$ | $1110\ 0001 = 0xE1$    |
| $0x66 = 0110\ 0110$ | $0011\ 0000 = 0x30$ | $0001\ 1001 = 0x19$ | $0001\ 1001 = 0x19$    |

### Problem 2.17

| Hexadecimal | Binary   | $B2U_4(x)$                   | $B2T_4(x)$                    |
|-------------|----------|------------------------------|-------------------------------|
| $0xE$       | $[1110]$ | $2^3 + 2^2 + 2^1 = 14$       | $-2^3 + 2^2 + 2^1 = -2$       |
| $0x0$       | $[0000]$ | 0                            | 0                             |
| $0x5$       | $[0101]$ | $2^2 + 2^0 = 5$              | $2^2 + 2^0 = 5$               |
| $0x8$       | $[1000]$ | $2^3 = 8$                    | $-2^3 = -8$                   |
| $0xD$       | $[1101]$ | $2^3 + 2^2 + 2^0 = 13$       | $-2^3 + 2^2 + 2^0 = -3$       |
| $0xF$       | $[1111]$ | $2^3 + 2^2 + 2^1 + 2^0 = 15$ | $-2^3 + 2^2 + 2^1 + 2^0 = -1$ |

### Problem 2.18

- A.  $0x2e0 = 736$
- B.  $-0x58 = -88$
- C.  $0x28 = 40$
- D.  $-0x30 = -48$
- E.  $0x78 = 120$
- F.  $0x88 = 136$
- G.  $0x1f8 = 504$
- H.  $0xc0 = 192$
- I.  $-0x48 = -72$

### Problem 2.19

| $x$ | $T2U_4(x)$                   |
|-----|------------------------------|
| -8  | 8                            |
| -3  | $2^3 + 2^2 + 2^0 = 13$       |
| -2  | $2^3 + 2^2 + 2^1 = 14$       |
| -1  | $2^3 + 2^2 + 2^1 + 2^0 = 15$ |
| 0   | 0                            |
| 5   | 5                            |

### Problem 2.20

Equation 2.5 can be used to solve the previous problem. Since  $\omega = 4$ , we need to add  $2^4 = 16$  to all negative numbers in Two's Complement. For example,  $-8 + 16 = 8$  and  $-1 + 16 = 15$ . Positive numbers (and zero) stay the same.

### Problem 2.21

| Expression                       | Type     | Evaluation |
|----------------------------------|----------|------------|
| $-2147483647 - 1 == 2147483648U$ | Unsigned | 1          |
| $-2147483647 - 1 < 2147483647$   | Signed   | 1          |
| $-2147483647 - 1U < 2147483647$  | Unsigned | 0          |
| $-2147483647 - 1 < -2147483647$  | Signed   | 1          |
| $-2147483647 - 1U < -2147483647$ | Unsigned | 1          |

### Problem 2.22

- A.  $[1011] = -2^3 + 2^1 + 2^0 = -5$   
 B.  $[11011] = -2^4 + 2^3 + 2^1 + 2^0 = -5$   
 C.  $[111011] = -2^5 + 2^4 + 2^3 + 2^1 + 2^0 = -5$

### Problem 2.23

| $w$        | $\text{fun1}(w)$ | $\text{fun2}(w)$ |
|------------|------------------|------------------|
| 0x00000076 | 0x00000076       | 0x00000076       |
| 0x87654321 | 0x00000021       | 0x00000021       |
| 0x000000C9 | 0x000000C9       | 0xFFFFF9C9       |
| 0xEDCBA987 | 0x00000087       | 0xFFFFF087       |

`fun1` keeps only the least significant byte and sets the other three to all zeroes, resulting in a value between 0 and 255. `fun2` also extracts the least significant byte, but it performs sign extension instead of zero extension, which results in a value between -128 and 127.

### Problem 2.24

| Hex      |           | Unsigned |           | Two's complement |           |
|----------|-----------|----------|-----------|------------------|-----------|
| Original | Truncated | Original | Truncated | Original         | Truncated |
| 0        | 0         | 0        | 0         | 0                | 0         |
| 2        | 2         | 2        | 2         | 2                | 2         |
| 9        | 1         | 9        | 1         | -7               | 1         |
| B        | 3         | 11       | 3         | -5               | 3         |
| F        | 7         | 15       | 7         | -1               | -1        |

We can use the equations to verify these results. For example, in hex F truncates to 7, in unsigned  $B2U_4(1111) \bmod 2^3 = 7$  and in two's complement  $U2T_3(B2U_4(1111) \bmod 2^3) = -1$ .

### Problem 2.25

Because `length` is unsigned the expression  $0 - 1$  evaluates to `UMax`. The comparison has an unsigned integer on one side, which means the other side will also be treated as unsigned. Of course every unsigned number is  $\leq \text{UMax}$  and so we try to access invalid array elements.

We can fix it by changing the condition to `i < length` or changing `length` to a signed integer.

### Problem 2.26

A. The function returns wrong results in case `t` is longer than `s`.

B. The problem is that `strlen` returns a `size_t` which is unsigned. When calculating `strlen(s) - strlen(t)` where `t` is longer than `s` unsigned arithmetic is used, resulting in a number close to `UMax` instead of a negative number. This is obviously greater than 0 so the function incorrectly says that `s` is longer.

C. We can fix it by changing the condition to `strlen(s) > strlen(t)`.

### Problem 2.27

```
/* Determine whether arguments can be added without overflow */
int uadd_ok(unsigned x, unsigned y) {
    return x + y >= x;
}
```

### Problem 2.28

| $x$ |         | $-\omega^u x$ |     |
|-----|---------|---------------|-----|
| Hex | Decimal | Decimal       | Hex |
| 0   | 0       | 0             | 0   |
| 5   | 5       | 11            | B   |
| 8   | 8       | 8             | 8   |
| D   | 13      | 3             | 3   |
| F   | 15      | 1             | 1   |

### Problem 2.29

| $x$            | $y$            | $x + y$         | $x + \frac{t}{5} y$ | Case |
|----------------|----------------|-----------------|---------------------|------|
| -12<br>[10100] | -15<br>[10001] | -27<br>[100101] | 5<br>[00101]        | 1    |
| -8<br>[11000]  | -8<br>[11000]  | -16<br>[110000] | -16<br>[10000]      | 2    |
| -9<br>[10111]  | 8<br>[01000]   | -1<br>[111111]  | -1<br>[11111]       | 2    |
| 2<br>[00010]   | 5<br>[00101]   | 7<br>[000111]   | 7<br>[00111]        | 3    |
| 12<br>[01100]  | 4<br>[00100]   | 16<br>[010000]  | -16<br>[10000]      | 4    |

### Problem 2.30

```

/* Determine whether arguments can be added without overflow */
int tadd_ok(int x, int y) {
    int sum = x + y;

    if (x > 0 && y > 0) {
        return sum > 0;
    }

    if (x < 0 && y < 0) {
        return sum < 0;
    }

    return 1;
}

```

### Problem 2.31

sum  $-x$  can overflow again, since it's another two's complement addition. For example, if  $x$  and  $y$  are large positive numbers whose sum overflows to a negative number, then sum  $-x$  will cause a negative overflow "wrapping back around" to  $y$ . So this check will not detect the overflow.

### Problem 2.32

The function will be incorrect for  $y = \text{TMin}_\omega$ . This is because the two's complement representation is not symmetric.  $-y = -\text{TMin}_\omega = \text{TMin}_\omega$  causes an overflow possibly resulting in an incorrect return value.

### Problem 2.33

| $x$ |         | $-\frac{t}{4}x$ |     |
|-----|---------|-----------------|-----|
| Hex | Decimal | Decimal         | Hex |
| 0   | 0       | 0               | 0   |
| 5   | 5       | -5              | B   |
| 8   | -8      | -8              | 8   |
| D   | -3      | 3               | 3   |
| F   | -1      | 1               | 1   |

The bit patterns for two's complement and unsigned negation are the same.

### Problem 2.34

| Mode             | $x$          | $y$          | $x \cdot y$     | Truncated $x \cdot y$ |
|------------------|--------------|--------------|-----------------|-----------------------|
| Unsigned         | $4 = [100]$  | $5 = [101]$  | $20 = [010100]$ | $4 = [100]$           |
| Two's complement | $-4 = [100]$ | $-3 = [101]$ | $12 = [001100]$ | $-4 = [100]$          |
| Unsigned         | $2 = [010]$  | $7 = [111]$  | $14 = [001110]$ | $6 = [110]$           |
| Two's complement | $2 = [010]$  | $-1 = [111]$ | $-2 = [111110]$ | $-2 = [110]$          |
| Unsigned         | $6 = [110]$  | $6 = [110]$  | $36 = [100100]$ | $4 = [100]$           |
| Two's complement | $-2 = [110]$ | $-2 = [110]$ | $4 = [000100]$  | $-4 = [100]$          |

### Problem 2.35

1. Let  $t = u + p_{\omega-1}$  where  $u$  is the two's complement number represented by the  $\omega$  upper bits of the  $2\omega$ -bit representation of  $x \cdot y$ . Since  $p_{\omega-1}$  is either 0 or 1, there are two possibilities for  $t$  to equal 0.

1. If  $p_{\omega-1} = 0$  then it must be that  $u = 0$  (upper  $\omega$  bits are all 0s).
2. If  $p_{\omega-1} = 1$  then it must be that  $u = -1$  (upper  $\omega$  bits are all 1s).

So  $t = 0$  if the upper  $\omega + 1$  bits are all 0s or all 1s. These are exactly the cases where the multiplication does not overflow. All other cases do overflow.

This means we can write  $x \cdot y = p + t2^\omega$  which overflows iff  $t \neq 0$ .

2. To show that  $p$  can be written in the form  $p = x \cdot q + r$ , where  $|r| < |q|$  we consider integer division. Dividing  $p$  by nonzero  $x$  gives a quotient  $q$  and remainder  $r$ , such that  $|r| < |q|$ .

3. By plugging in we get  $x \cdot y = x \cdot q + r + t2^\omega$ . If  $r + t2^\omega = 0$  then  $q = y$ . Since  $|r| < |q| < 2^\omega$  this can only hold if  $r = t = 0$ .

### Problem 2.36

```
/* Determine whether arguments can be multiplied without overflow */
int tmult_ok(int x, int y) {
    int64_t prod = ((int64_t)x) * y;
    int64_t upper = prod >> 31;
    // if the upper 33 bits are all 1s or 0s the number fits into 32 bits
    return upper == 0 || upper == -1;
}
```

### Problem 2.37

A. The new code does not improve the situation since the 64-bit number will be truncated to 32 bits when passed to `malloc`. This truncation is the same that also happens when the multiplication overflows.

B. Check the multiplication for overflow (by one of the previous methods) and if it overflows immediately abort and don't allocate any memory.

### Problem 2.38

A single LEA instruction can compute the following multiples:

| $k$ | $b$ | $(a \ll k) + b$   |
|-----|-----|-------------------|
| 0   | 0   | $(2^0 + 0)a = 1a$ |
| 0   | a   | $(2^0 + 1)a = 2a$ |
| 1   | 0   | $(2^1 + 0)a = 2a$ |
| 1   | a   | $(2^1 + 1)a = 3a$ |
| 2   | 0   | $(2^2 + 0)a = 4a$ |
| 2   | a   | $(2^2 + 1)a = 5a$ |
| 3   | 0   | $(2^3 + 0)a = 8a$ |
| 3   | a   | $(2^3 + 1)a = 9a$ |

### Problem 2.39

In this case the expression simplifies to  $-(x \ll m)$ . This is because shifting by  $n + 1 = \omega$  to the left results in 0 so we can ignore the first term.

### Problem 2.40

| $K$ | Shifts | Add/Subs | Expression                  |
|-----|--------|----------|-----------------------------|
| 6   | 2      | 1        | $(x \ll 2) + (x \ll 1)$     |
| 31  | 1      | 1        | $(x \ll 5) - x$             |
| -6  | 2      | 1        | $(x \ll 1) - (x \ll 3)$     |
| 55  | 2      | 2        | $(x \ll 6) - (x \ll 3) - x$ |

### Problem 2.41

Consider two cases:

- $m > 0$ : In this case form A requires  $n - m + 1$  shifts and  $n - m$  additions while form B requires 2 shifts and 1 subtraction. So if  $n = m$ , form A is favorable since it requires only 1 shift and 0 additions which is less than the constant numbers required for form B. If  $n = m + 1$ , form A takes 2 shifts and 1 addition, so either form is equally efficient in this case. If  $n > m + 1$  form A takes  $n - m + 1 > 2$  shifts and  $n - m > 1$  additions so form B is favorable in this case.
- $m = 0$ : In this case the last bit does not cause a shift so form A requires  $n$  shifts and  $n$  additions, while form B takes 1 shift and 1 subtraction. The same three cases apply in the same way here, so the above analysis extends to this case as well.

### Problem 2.42

```
int div16(int x) {  
    int bias = (x >> 31) & 15;  
    return (x + bias) >> 4;  
}
```

### Problem 2.43

$x$  is shifted to the left by 5 which is equivalent to multiplying by  $2^5 = 32$ . Then, one  $x$  is subtracted from it, so we end up with  $31x$  and  $M = 31$ .

If  $y$  is negative, a bias of  $7 = 8 - 1$  is added. Then  $y$  is shifted right arithmetically by 3 which is equivalent to dividing by  $2^3 = 8$ . This means that  $N = 8$ .

### Problem 2.44

A. `false` for  $x = -2^{31}$  which is `INT32_MIN`.  $x$  is obviously not greater than 0, and  $x - 1$  overflows to `INT32_MAX` which is not lower than 0.

B. Always `true`. The expressions are connected by `OR` so both parts would need to evaluate to 0.

Let  $x_2$  be  $x$ 's third bit from the right. For the first part to evaluate to 0 we need  $x_2 = 1$ . For the second part,  $x_2$  will become the sign so it needs to be 0 to represent a positive number. This is obviously not possible at the same time so at least one part always evaluates to 1.



| Bits    | $e$ | $E$ | $2^E$ | $f$           | $M$           | $2^E \times M$ | $V$           | Decimal |
|---------|-----|-----|-------|---------------|---------------|----------------|---------------|---------|
| 0 00 11 | 0   | 0   | 1     | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$  | $\frac{3}{4}$ | 0.75    |
| 0 01 00 | 1   | 0   | 1     | $\frac{0}{4}$ | $\frac{4}{4}$ | $\frac{4}{4}$  | 1             | 1.0     |
| 0 01 01 | 1   | 0   | 1     | $\frac{1}{4}$ | $\frac{5}{4}$ | $\frac{5}{4}$  | $\frac{5}{4}$ | 1.25    |
| 0 01 10 | 1   | 0   | 1     | $\frac{2}{4}$ | $\frac{6}{4}$ | $\frac{6}{4}$  | $\frac{3}{2}$ | 1.5     |
| 0 01 11 | 1   | 0   | 1     | $\frac{3}{4}$ | $\frac{7}{4}$ | $\frac{7}{4}$  | $\frac{7}{4}$ | 1.75    |
| 0 10 00 | 2   | 1   | 2     | $\frac{0}{4}$ | $\frac{4}{4}$ | $\frac{8}{4}$  | 2             | 2.0     |
| 0 10 01 | 2   | 1   | 2     | $\frac{1}{4}$ | $\frac{5}{4}$ | $\frac{10}{4}$ | $\frac{5}{2}$ | 2.5     |
| 0 10 10 | 2   | 1   | 2     | $\frac{2}{4}$ | $\frac{6}{4}$ | $\frac{12}{4}$ | 3             | 3.0     |
| 0 10 11 | 2   | 1   | 2     | $\frac{3}{4}$ | $\frac{7}{4}$ | $\frac{14}{4}$ | $\frac{7}{2}$ | 3.5     |
| 0 11 00 | -   | -   | -     | -             | -             | -              | $\infty$      | -       |
| 0 11 01 | -   | -   | -     | -             | -             | -              | NaN           | -       |
| 0 11 10 | -   | -   | -     | -             | -             | -              | NaN           | -       |
| 0 11 11 | -   | -   | -     | -             | -             | -              | NaN           | -       |

### Problem 2.48

3'510'593 in binary is 11 0101 1001 0001 0100 0001 which equals  $1.101011001000101000001 \times 2^{21}$ .  
So the biased exponent is  $21 + 2^7 - 1 = 148$ .

The complete single-precision floating-point representation is:

0 10010100 10101100100010100000100 or 0x4A564504

Comparing the two we see that part of it overlaps:

```
000000000001101011001000101000001
*****
01001010010101100100010100000100
```

### Problem 2.49

A.  $2^{n+1} + 1$

B.  $2^{24} + 1 = 16'777'217$

### Problem 2.50

| Exact      |                | Rounded  |                |
|------------|----------------|----------|----------------|
| Binary     | Decimal        | Binary   | Decimal        |
| $10.010_2$ | $2\frac{1}{4}$ | $10.0_2$ | 2              |
| $10.011_2$ | $2\frac{3}{8}$ | $10.1_2$ | $2\frac{1}{2}$ |
| $10.110_2$ | $2\frac{3}{4}$ | $11.0_2$ | 3              |
| $11.001_2$ | $3\frac{1}{8}$ | $11.0_2$ | 3              |

### Problem 2.51

A. 0.00011001100110011001101

B.

```
0.00011001100110011001101
- 0.0001100110011001100110011 0011 0011...
-----
0.000000000000000000000000[1100]
```

This is equal to  $\frac{1}{10} \times 2^{-22} \approx 2.38 \times 10^{-8}$ .

C. After 100 hours the clock is ahead by  $2.38 \times 10^{-8} \cdot 100 \cdot 60 \cdot 60 \cdot 10 \approx 0.086$  seconds.

D.  $2000\text{m/s} \cdot 0.086\text{s} \approx 172\text{m}$

### Problem 2.52

| Format A |                 | Format B |                |
|----------|-----------------|----------|----------------|
| Bits     | Value           | Bits     | Value          |
| 011 0000 | 1               | 0111 000 | 1              |
| 101 1110 | $\frac{15}{2}$  | 1001 111 | $\frac{15}{2}$ |
| 010 1001 | $\frac{25}{32}$ | 0110 100 | $\frac{3}{4}$  |
| 110 1111 | $\frac{31}{2}$  | 1011 000 | 16             |
| 000 0001 | $\frac{1}{64}$  | 0001 000 | $\frac{1}{64}$ |

### Problem 2.53

```
#define POS_INFINITY 1e400 // overflows
#define NEG_INFINITY (-POS_INFINITY)
#define NEG_ZERO (1.0/NEG_INFINITY)
```

### Problem 2.54

A. Always `true`. `double` has enough range to represent all `int` values and it does not need to round. This also means no rounding when casting back.

B. `false` for  $2^{24} + 1$ . It is not true for all numbers with 24 or more significant bits, since `float` uses 23 bits for representing the fraction, which means these values get rounded and lose precision.

C. `false` for  $2^{24} + 1$ . The same reasoning as above applies here.

D. Always `true`. `double` has enough range and precision to exactly represent any `float`.

E. Always `true`. Negating the `float` only flips the sign bit and loses no information. So flipping it back results in the original value.

F. Always `true`. The integers get converted to `float` first.

G. Always `true`. If the multiplication overflows it results in  $\infty > 0$ .

H. `false` for  $f = 10^{20}$  and  $d = 1.0$ . All values are promoted to `double` but even then the precision is not enough to represent  $10^{20} + 1.0$ . This is because  $\log_2 10^{20} > 66$  and `double` has only 52 bits for representing the fraction. So when we set the least significant bit to 1 we would need about 66 bits to store the precise number. This means that adding 1.0 will have no effect on the very large number.

### Problem 2.58

```
int is_little_endian() {
    // set only least significant byte to 0x01
    uint16_t x = 1;
    // char pointer reads only first byte, so it will be:
    // 0x00 on big endian systems
    // 0x01 on little endian systems
    return *(char *)&x;
}
```

### Problem 2.59

```
int main() {
    int x = 0x89ABCDEF;
    int y = 0x76543210;
    int mask = 0xFF;
    printf("%d\n", ((x & mask) | (y & ~mask)) == 0x765432EF);
}
```

### Problem 2.60

```
unsigned replace_byte(unsigned x, int i, unsigned char b) {
    unsigned mask = ~(0xFF << (i << 3));
    return (x & mask) | (b << (i << 3));
}

int main() {
    printf("%d\n", replace_byte(0x12345678, 2, 0xAB) == 0x12AB5678);
    printf("%d\n", replace_byte(0x12345678, 0, 0xAB) == 0x123456AB);
}
```

### Problem 2.61

```
int A = !!x;
int B = !!~x;
int C = !(x & 0xFF);
int D = !!~(x >> ((sizeof(int) - 1) << 3));
```

### Problem 2.62

```
int int_shifts_are_arithmetic() {
    return (-1 >> 1) == -1;
}
```

### Problem 2.63

```
unsigned srl(unsigned x, int k) {
    /* Perform shift arithmetically */
    unsigned xsra = (int)x >> k;

    int w = sizeof(int) << 3;
    int mask = ~(-1 << (w - k));
    return xsra & mask;
}

int sra(int x, int k) {
    /* Perform shift logically */
    int xsrl = (unsigned) x >> k;

    int w = sizeof(int) << 3;
    int mask = -1 << (w - k);
    int sign = -(x & (1 << (w - 1)));
    return xsrl | (mask & sign);
}
```

### Problem 2.64

```
int any_odd_one(unsigned x) {
    int mask = 0b101010101010101010101010101010;
    return !(x & mask);
}
```

### Problem 2.65

```
int odd_ones(unsigned x) {
    x ^= x >> 16;
    x ^= x >> 8;
    x ^= x >> 4;
    x ^= x >> 2;
    x ^= x >> 1;
    return x & 1;
}
```

### Problem 2.66

```
int leftmost_one(unsigned x) {
    // Spread leftmost 1 to the right
    x |= x >> 1;
    x |= x >> 2;
    x |= x >> 4;
    x |= x >> 8;
    x |= x >> 16;
    return x ^ (x >> 1);
}
```

### Problem 2.67

Shifting by an amount  $\geq$  the width of the type is undefined for signed integers.

```
int int_size_is(unsigned w) {
    unsigned set_msb = 1U << (w - 1);
    unsigned beyond_msb = set_msb << 1;
    return set_msb && !beyond_msb;
}
```

### Problem 2.68

```
int lower_one_mask(int n) {
    int w = sizeof(int) << 3;
    return ~0U >> (w - n);
}
```

### Problem 2.69

```
unsigned rotate_left(unsigned x, int n) {
    unsigned w = sizeof(unsigned) << 3;
    unsigned rotated_bits = x >> (w - n - 1) >> 1;
    return (x << n) | rotated_bits;
}
```

### Problem 2.70

```
int fits_bits(int x, int n) {
    int w = sizeof(int) << 3;
    int shifted = x << (w - n) >> (w - n);
    return shifted == x;
}
```

### Problem 2.71

- A. The upper three bytes are always all zeroes, which means we can't represent negative numbers.
- B. Correct implementation:

```
typedef unsigned packed_t;

int xbyte(packed_t word, int bytenum) {
    // cut off unwanted bytes to the left
    int shifted = word << ((3 - bytenum) << 3);
    // cut off unwanted bytes to the right
    // number will be sign extended because type is int
    return shifted >> 24;
}
```

### Problem 2.72

- A. Because `sizeof` returns a `size_t` (which is unsigned), the result of `maxbytes - sizeof(val)` is also unsigned and thus always  $\geq 0$ .
- B. Correct implementation:

```
void copy_int(int val, void *buf, int maxbytes) {
    if (maxbytes >= sizeof(val))
        memcpy(buf, (void *)&val, sizeof(val));
}
```

### Problem 2.73

```
int saturating_add(int x, int y) {
    int sum = x + y;
    int offset = (sizeof(int) << 3) - 1;

    // create masks with all 0s or all 1s
    int x_neg = x >> offset;
    int y_neg = y >> offset;
    int sum_neg = sum >> offset;

    int pos_overflow = ~x_neg & ~y_neg & sum_neg;
    int neg_overflow = x_neg & y_neg & ~sum_neg;

    // use masks to choose between sum and bounds
    return (INT32_MAX & pos_overflow) | (INT32_MIN & neg_overflow) |
        (sum & ~(pos_overflow | neg_overflow));
}
```

### Problem 2.74

```
int tsub_ok(int x, int y) {
    int offset = (sizeof(int) << 3) - 1;
    int x_neg = x >> offset;
    int y_neg = y >> offset;
    int diff_neg = (x - y) >> offset;

    int pos_overflow = ~x_neg & y_neg & diff_neg;
    int neg_overflow = x_neg & ~y_neg & ~diff_neg;
    return !(pos_overflow | neg_overflow);
}
```

### Problem 2.75

```
unsigned unsigned_high_prod(unsigned x, unsigned y) {
    unsigned prod = signed_high_prod(x, y);
    unsigned offset = (sizeof(unsigned) << 3) - 1;
    unsigned x_sign = x >> offset;
    unsigned y_sign = y >> offset;
    return prod + x_sign * y + y_sign * x;
}
```

## Problem 2.76

```
void *calloc(size_t nmemb, size_t size) {
    if (nmemb == 0 || size == 0) {
        return NULL;
    }

    size_t total_size = nmemb * size;

    if (total_size / size != nmemb) {
        // Overflow occurred
        return NULL;
    }

    void *ptr = malloc(total_size);

    if (ptr == NULL) {
        // Allocation failed
        return NULL;
    }

    memset(ptr, 0, total_size);
    return ptr;
}
```

## Problem 2.77

```
int times_17 = (x << 4) + x;
int times_neg7 = x - (x << 3);
int times_60 = (x << 6) - (x << 2);
int times_neg112 = (x << 4) - (x << 7);
```

## Problem 2.78

```
int divide_power(int x, int k) {
    int w = sizeof(int) << 3;
    int mask = x >> (w - 1);
    int bias = mask & ((1 << k) - 1);
    return (x + bias) >> k;
}
```

### Problem 2.79

```
int mul3div4(int x) {
    int mul3 = (x << 1) + x;
    int w = sizeof(int) << 3;
    int mask = mul3 >> (w - 1);
    int bias = mask & 3;
    return (mul3 + bias) >> 2;
}
```

### Problem 2.80

```
int threefourths(int x) {
    // calculate result for higher bits that are not relevant for rounding
    int high_div4 = x >> 2;
    int high_result = (high_div4 << 1) + high_div4;

    // for the remainder we can safely multiply first since it cannot overflow
    int rem = x - (high_div4 << 2);
    int rem_mul3 = (rem << 1) + rem;
    int mask = x >> ((sizeof(int) << 3) - 1);
    int bias = mask & 3;
    int low_result = (rem_mul3 + bias) >> 2;

    return high_result + low_result;
}
```

### Problem 2.81

```
int main() {
    // example values
    int k = 9;
    int j = 5;

    int A = -1 << k;
    int B = ((1 << k) - 1) << j;
}
```

### Problem 2.82

- A. `false` for  $x = -2^{31}$  which is  $\text{TMin}_{32}$  because  $-\text{TMin} = \text{TMin}$  (since it overflows).
- B. Always `true`. If any overflows happen they will be the same on both sides. (ring properties of two's complement arithmetic)

C. Always `true`. `~x` is equal to `-x - 1`. So we can write it as

`-x - 1 - y - 1 + 1 = -(x+y) - 1` which simplifies to `-x - y - 1 = -x - y - 1`.

D. Always `true`. By negating we turn  $y - x$  into  $x - y$ . Regardless of the negation, both sides are treated as unsigned for the comparison. And unsigned and two's complement arithmetic are the same on the bit level.

E. Always `true`. This basically sets the 2 lower bits to 0. These bits have a positive weight regardless of the sign bit, so setting them to 0 can not make the number larger.

### Problem 2.83

A. Let  $V$  be the value of the string. Shifting the binary point  $k$  to the right results in  $y.yyyy\dots$  which equals both  $V + Y$  and  $V \times 2^k$ . These can now be equated:

$$\begin{aligned} V + Y &= V \times 2^k \\ Y &= V \times 2^k - V \\ Y &= V(2^k - 1) \\ V &= \frac{Y}{2^k - 1} \end{aligned}$$

B.

(a)  $V = \frac{5}{2^3 - 1} = \frac{5}{7}$

(b)  $V = \frac{6}{2^4 - 1} = \frac{2}{5}$

(c)  $V = \frac{19}{2^6 - 1} = \frac{19}{63}$

### Problem 2.84

```
return ((ux << 1) == 0 && (uy << 1) == 0) // +0 and -0 are considered equal
    || (!sx && !sy && ux <= uy)           // both positive
    || (sx && sy && ux >= uy)             // both negative
    || (sx > sy);                         // different signs
```

### Problem 2.85

A. The number 7.0 will have exponent  $E = 2$ , significand  $M = 1.11_2 = \frac{7}{4}$ , fraction  $f = 0.11_2 = \frac{3}{4}$  and value  $V = 7$ . The exponent will be represented as `10...01` (bias is  $2^{k-1} - 1 = 01\dots1_2$ ) and the fraction as `110...0`.

B. The largest odd integer that can be represented exactly will have exponent  $E = n$ , significand  $M = 1.1\dots1 = 2 - \frac{1}{2^n}$ , fraction  $f = 0.1\dots1_2 = 1 - \frac{1}{2^n}$  and value  $V = 2^{n+1} - 1$ . The exponent will be the binary representation of  $n + 2^{k-1} - 1$  and the fraction will be represented as `1...1`.

C. The smallest positive normalized value has exponent  $2 - 2^{k-1}$  and fraction 0, giving a value of  $1.0 \times 2^{2-2^{k-1}}$ . The reciprocal of this is  $V = 2^{2^{k-1}-2}$ . It will have exponent  $E = 2^{k-1} - 2$ , significand  $M = 1$  and fraction  $f = 0$ . The biased exponent will be  $2^{k-1} - 2 - 2^{k-1} - 1 = -3$  represented by `1...101` and the fraction by `0...0`.

### Problem 2.86

| Description                    | Extended precision                  |                           |
|--------------------------------|-------------------------------------|---------------------------|
|                                | Value                               | Decimal                   |
| Smallest positive denormalized | $2^{-63} \times 2^{2-2^{14}}$       | $2.645 \times 10^{-4951}$ |
| Smallest positive normalized   | $2^{2-2^{14}}$                      | $3.362 \times 10^{-4932}$ |
| Largest normalized             | $(2 - 2^{-63}) \times 2^{2^{14}-1}$ | $1.190 \times 10^{4932}$  |

### Problem 2.87

| Description                                      | Hex                 | $M$                 | $E$ | $V$                   | $D$          |
|--------------------------------------------------|---------------------|---------------------|-----|-----------------------|--------------|
| -0                                               | <code>0x8000</code> | 0                   | -14 | -0                    | -0.0         |
| Smallest value > 2                               | <code>0x4001</code> | $\frac{1025}{1024}$ | 1   | $1025 \times 2^{-9}$  | 2.001953125  |
| 512                                              | <code>0x6000</code> | 1                   | 9   | 512                   | 512.0        |
| Largest denormalized                             | <code>0x03FF</code> | $\frac{1023}{1024}$ | -14 | $1023 \times 2^{-24}$ | 0.0000609756 |
| $-\infty$                                        | <code>0xFC00</code> | -                   | -   | $-\infty$             | $-\infty$    |
| Number with hex representation <code>3BB0</code> | <code>0x3BB0</code> | $\frac{123}{64}$    | -1  | $123 \times 2^{-7}$   | 0.9609375    |

### Problem 2.88

| Format A                 |                    | Format B                 |                   |
|--------------------------|--------------------|--------------------------|-------------------|
| Bits                     | Value              | Bits                     | Value             |
| <code>1 01111 001</code> | $\frac{-9}{8}$     | <code>1 0111 0010</code> | $\frac{-9}{8}$    |
| <code>0 10110 011</code> | 176                | <code>0 1110 0110</code> | 176               |
| <code>1 00111 010</code> | $\frac{-5}{1024}$  | <code>1 0000 0101</code> | $\frac{-5}{1024}$ |
| <code>0 00000 111</code> | $\frac{7}{2^{17}}$ | <code>0 0000 0001</code> | $\frac{1}{1024}$  |
| <code>1 11100 000</code> | $-2^{13}$          | <code>1 1110 1111</code> | -248              |
| <code>0 10111 100</code> | 384                | <code>0 1111 0000</code> | $\infty$          |

### Problem 2.89

- A. Always `true`. `float` cannot represent every 32-bit integer value in full precision, but `double` can, so they will be rounded in the same way.
- B. `false` for  $x = 0$  and  $y = \text{TMin}_{32}$ . The integer result will overflow before it is cast to `double` while the operation with `double` s has a wider range and does not overflow.
- C. Always `true`. The result of adding two floats in the 32-bit integer range cannot overflow.
- D. (Apparently not always true, cannot reproduce solution tho)
- E. `false` for  $x = 1$  and  $z = 0$ . Dividing by 0 results in `NaN`.

### Problem 2.90

```
float fpwr2(int x) {
    /* Result exponent and fraction */
    unsigned exp, frac;
    unsigned u;

    if (exp < -126-23) {
        /* Too small. Return 0.0 */
        exp = 0;
        frac = 0;
    } else if (exp < -126) {
        /* Demormalized result */
        exp = 0;
        frac = 1 << (exp + 126 + 23);
    } else if (exp <= 127) {
        /* Normalized result */
        exp = exp + 127;
        frac = 0;
    } else {
        /* Too big. Return +inf */
        exp = 0xFF;
        frac = 0;
    }

    /* Pack exp and frac into 32 bits */
    u = exp << 23 | frac;
    /* Return as float */
    return u2f(u);
}
```

### Problem 2.91

- A. Binary representation: `0 10000000 10010010000111111011011`. So the exponent is  $128 - 127 = 1$  and the fractional number is  $11.0010010000111111011011_2$ .

B.  $\frac{22}{7} = 3\frac{1}{7} = 11.001001001001001\dots_2$

C. The two approximations diverge starting from the 9th bit after the binary point.

### Problem 2.92

```
typedef unsigned float_bits;

float_bits float_negate(float_bits f) {
    unsigned exp = f >> 23 & 0xFF;
    unsigned frac = f & 0x7FFFFFFF;

    if (exp == 0xFF && frac != 0) {
        // NaN, return as is
        return f;
    };

    unsigned mask = 1 << 31;
    return f ^ mask;
}
```

### Problem 2.93

```
typedef unsigned float_bits;

float_bits float_absval(float_bits f) {
    unsigned exp = f >> 23 & 0xFF;
    unsigned frac = f & 0x7FFFFFFF;

    if (exp == 0xFF && frac != 0) {
        // NaN, return as is
        return f;
    };

    unsigned mask = (1U << 31) - 1;
    return f & mask;
}
```

## Problem 2.94

```
typedef unsigned float_bits;

float_bits float_twice(float_bits f) {
    unsigned sign = f >> 31;
    unsigned exp = (f >> 23) & 0xFF;
    unsigned frac = f & 0x7FFFFFFF;

    if (exp == 0xFF) {
        // NaN, +-Infinity, return as is
        return f;
    };

    if (exp == 0) {
        // Denormalized number
        frac <<= 1;

        if (frac > 0x7FFFFFFF) {
            // normalize
            exp = 1;
            frac &= 0x7FFFFFFF;
        }
        return sign << 31 | (exp << 23) | frac;
    }

    // normalized number
    exp += 1;

    if (exp == 0xFF) {
        // overflow to infinity
        frac = 0;
    }
    return sign << 31 | (exp << 23) | frac;
}
```

## Problem 2.95

```
typedef unsigned float_bits;

float_bits float_half(float_bits f) {
    unsigned sign = f >> 31;
    unsigned exp = (f >> 23) & 0xFF;
    unsigned frac = f & 0x7FFFFFFF;

    if (exp == 0xFF) {
        // NaN, +-Infinity, return as is
        return f;
    };

    // round to even
    unsigned roundup = (frac & 0x3) == 3;

    if (exp == 0) {
        // Denormalized number
        frac >>= 1;
        frac += roundup;
        return sign << 31 | (exp << 23) | frac;
    }

    // normalized number
    exp -= 1;

    if (exp == 0) {
        // denormalize
        frac |= 0x800000; // add implicit leading 1
        frac >>= 1;
        frac += roundup;
    }

    return sign << 31 | (exp << 23) | frac;
}
```

## Problem 2.96

```
typedef unsigned float_bits;

int float_f2i(float_bits f) {
    unsigned sign = f >> 31;
    unsigned exp = (f >> 23) & 0xFF;
    unsigned frac = f & 0x7FFFFFFF;

    if (exp == 0xFF) {
        // NaN or infinity
        return 0x80000000;
    }

    if (exp == 0) {
        // Denormalized number
        return 0;
    }

    // Normalized number
    int unbiased_exp = exp - 127;

    if (unbiased_exp < 0) {
        // Absolute value less than 1
        return 0;
    }

    if (unbiased_exp >= 31) {
        // Overflow
        return 0x80000000;
    }

    if (unbiased_exp < 23) {
        frac >>= 23 - unbiased_exp;
    } else {
        frac <<= unbiased_exp - 23;
    }

    return (sign ? -1 : 1) * ((1 << unbiased_exp) + frac);
}
```

## Problem 2.97

```
typedef unsigned float_bits;

float_bits float_i2f(int i) {
    unsigned sign = i & (1U << 31);
    unsigned abs = i < 0 ? -i : i;

    if (abs == 0) {
        return 0;
    }

    // normalized number
    unsigned exp = 127 + 31;

    // shift left until bit 31 is set (will become implicit leading 1)
    while ((abs & (1U << 31)) == 0) {
        abs <<= 1;
        exp--;
    }

    unsigned frac = (abs >> 8) & 0x7FFFFFFF;
    unsigned residue = abs & 0xFF;

    // round to even
    if (residue > 0x80 || residue == 0x80 && (frac & 1)) {
        frac++;

        if (frac > 0x7FFFFFFF) {
            // renormalize
            frac &= 0x7FFFFFFF;
            frac >>= 1;
            exp++;
        }
    }

    return sign | (exp << 23) | frac;
}
```

## Chapter 3 Machine-Level Representation of Programs

### Problem 3.1

| Operand | Value |
|---------|-------|
| %rax    | 0x100 |
| 0x104   | 0xAB  |
| \$0x108 | 0x108 |
| (%rax)  | 0xFF  |

| Operand        | Value |
|----------------|-------|
| 4(%rax)        | 0xAB  |
| 9(%rax,%rdx)   | 0x11  |
| 260(%rcx,%rdx) | 0x13  |
| 0xFC(,%rcx,4)  | 0xFF  |
| (%rax,%rdx,4)  | 0x11  |

### Problem 3.2

movl

movw

movb

movb

movq

movw

### Problem 3.3

| Instruction          | Mistake                                                       |
|----------------------|---------------------------------------------------------------|
| movb \$0xF, (%ebx)   | %ebx is a 32-bit register, memory addresses have to be 64 bit |
| movl %rax, (%rsp)    | %rax is 64 bit and thus must be used with movq                |
| movw (%rax), 4(%rsp) | both operands are memory locations                            |
| movb %al, %sl        | %sl is not a valid register                                   |
| movq %rax, \$0x123   | destination cannot be an immediate value                      |
| movl %eax, %rdx      | %rdx is 64 bit and thus must be used with movq                |
| movb %si, 8(%rbp)    | %si is 16 bit and thus must be used with movw                 |

### Problem 3.4

| src_t | dest_t | Instruction                            |
|-------|--------|----------------------------------------|
| long  | long   | movq (%rdi), %rax<br>movq %rax, (%rsi) |

| src_t         | dest_t        | Instruction                                      |
|---------------|---------------|--------------------------------------------------|
| char          | int           | <pre>movsbl (%rdi), %eax movl %eax, (%rsi)</pre> |
| char          | unsigned      | <pre>movsbl (%rdi), %eax movl %eax, (%rsi)</pre> |
| unsigned char | long          | <pre>movzbl (%rdi) %eax movq %rax (%rsi)</pre>   |
| int           | char          | <pre>movl (%rdi) %eax movb %al (%rsi)</pre>      |
| unsigned      | unsigned char | <pre>movl (%rdi) %eax movb %al (%rsi)</pre>      |
| char          | short         | <pre>movsbw (%rdi) %ax movw %ax (%rsi)</pre>     |

### Problem 3.5

```
void decode1(long *xp, long *yp, long *zp) {
    long x = *xp;
    long y = *yp;
    long z = *zp;
    *yp = x;
    *zp = y;
    *xp = z;
}
```

### Problem 3.6

| Instruction                            | Result       |
|----------------------------------------|--------------|
| <code>leaq 6(%rax), %rdx</code>        | $x + 6$      |
| <code>leaq (%rax,%rcx), %rdx</code>    | $x + y$      |
| <code>leaq (%rax,%rcx,4), %rdx</code>  | $x + 4y$     |
| <code>leaq 7(%rax,%rax,8), %rdx</code> | $9x + 7$     |
| <code>leaq 0xA(,%rcx,4), %rdx</code>   | $4y + 10$    |
| <code>leaq 9(%rax,%rcx,2), %rdx</code> | $x + 2y + 9$ |

### Problem 3.7

```
long t = 5 * x + 2 * y + 8 * z;
```

### Problem 3.8

| Instruction                 | Destination | Value |
|-----------------------------|-------------|-------|
| addq %rcx, (%rax)           | 0x100       | 0x100 |
| subq %rdx, 8(%rax)          | 0x108       | 0xA8  |
| imulq \$16, (%rax, %rdx, 8) | 0x118       | 0x110 |
| incq 16(%rax)               | 0x110       | 0x14  |
| decq %rcx                   | %rcx        | 0x0   |
| subq %rdx, %rax             | %rax        | 0xFD  |

### Problem 3.9

| C      | Assembly       |
|--------|----------------|
| x <= 4 | salq \$4, %rax |
| x >= n | sarq %cl, %rax |

### Problem 3.10

```
long arith2(long x, long y, long z) {
    long t1 = x | y;
    long t2 = t1 >> 3;
    long t3 = ~t2;
    long t4 = z - t3;
    return t4;
}
```

### Problem 3.11

- A. It sets the `rdx` register to all zeroes.
- B. `movq $0, %rdx` is a more straightforward way to express this.

C. The `xor` instruction only takes 3 bytes to encode, while `mov` takes 7 bytes. (Check using `echo 'movq $0, %rdx' | as -o /tmp/test.o | objdump -d /tmp/test.o`)

### Problem 3.12

```
void uremdiv(unsigned long x, unsigned long y,
            unsigned long *qp, unsigned long *rp)
x in %rdi, y in %rsi, qp in %rdx, rp in %rcx

uremdiv:
    movq %rdx, %r8      Copy qp
    movq %rdi, %rax      Move x to lower 8 bytes of dividend
    xorq %rdx, %rdx      Set upper 8 bytes of dividend to 0
    divq %rsi            Divide by y (unsigned)
    movq %rax, (%r8)      Store quotient at qp
    movq %rdx, (%rcx)     Store remainder at rp
    ret
```

### Problem 3.13

| Part | data_t                         | COMP |
|------|--------------------------------|------|
| A    | int                            | <    |
| B    | short                          | >=   |
| C    | unsigned char                  | <=   |
| D    | (unsigned) long / some pointer | !=   |

### Problem 3.14

| Part | data_t           | TEST |
|------|------------------|------|
| A    | long             | >=   |
| B    | (unsigned) short | ==   |
| C    | unsigned char    | >    |
| D    | int              | <=   |

### Problem 3.15

A. 4003fe

B. 400425

C. Address of `ja` is 400543 , address of `pop` is 400545 .

D. 400560

### Problem 3.16

A.

```
void cond(long a, long *p) {
    if (p == 0)
        goto done;
    if (*p >= a)
        goto done;
    *p = a;
done:
    return;
}
```

B. The assembly code contains two conditional branches because the condition in C consists of two parts. If the first part fails, the second part will be skipped (since they are connected by `&&` ).

### Problem 3.17

A.

```
long gotodiff_se(long x, long y) {
    long result;
    if (x < y)
        goto x_lt_y;
    ge_cnt++;
    result = x - y;
    return result;
x_lt_y:
    lt_cnt++;
    result = y - x;
    return result;
}
```

B. They are mostly equivalent. However, the first form is more convenient if there is no else case.

### Problem 3.18

```
long test(long x, long y, long z) {  
    long val = x + y + z;  
    if (x < -3) {  
        if (y < z)  
            val = x * y;  
        else  
            val = y * z;  
    } else if (x > 2)  
        val = x * z;  
    return val;  
}
```

### Problem 3.19

A.

$$T_{MP} = 2(T_{ran} - T_{OK})$$
$$T_{MP} = 2(31 - 16) = 30$$

B.

$$T_{OK} + T_{MP} = 16 + 30 = 46$$

### Problem 3.20

A. OP is division

B.

```
arith:  
    leaq    7(%rdi), %rax    store x + 7 in return register  
    testq   %rdi, %rdi      set condition codes based on x  
    cmovns  %rdi, %rax      set return value to x if x >= 0  
    sarq    $3, %rax        divide by 8 (arithmetic shift)  
    ret
```

For negative numbers we add a bias to round towards 0 instead of -inf. When dividing by  $2^k$  the bias is  $2^k - 1$ .

### Problem 3.21

```
long test(long x, long y) {
    long val = 8*x;
    if (y > 0) {
        if (x < y)
            val = y-x;
        else
            val = x&y;
    } else if (y <= -2)
        val = x+y;
    return val;
}
```

### Problem 3.22

- A. The maximum value of  $n$  for which we can represent  $n!$  with a 32-bit int is 12.
- B. The maximum value of  $n$  for which we can represent  $n!$  with a 64-bit long is 20.

This can be verified with the following code:

```
int main() {
    long result = 1;
    long prev_result = 1;
    int factor = 1;

    for (;;) {
        result *= ++factor;
        printf("%d %ld\n", factor, result);

        if (result / factor != prev_result) {
            printf("Overflow detected at factor %d\n", factor);
            break;
        }
        prev_result = result;
    }
}
```

### Problem 3.23

A.

| C Variable | Register                  |
|------------|---------------------------|
| x          | %rdi (initially) and %rax |
| y          | %rcx                      |
| n          | %rdx                      |

B. The pointer always points to `x` so `(*p)++` just increments `x`. This is then combined with `x += y` into a single `leaq` instruction.

C.

```
dw_loop:
    movq    %rdi, %rax           put x into the return register
    movq    %rdi, %rcx
    imulq   %rdi, %rcx           y = x*x
    leaq    (%rdi,%rdi), %rdx     n = 2*x
.L2:
    leaq    1(%rcx,%rax), %rax    x += y + 1
    subq    $1, %rdx             n--
    testq   %rdx, %rdx           test n
    jg      .L2                  loop if n > 0
    rep; ret                     return x
```

### Problem 3.24

```
long loop_while(long a, long b) {
    long result = 1;
    while (a < b) {
        result = result * (a+b);
        a = a + 1;
    }
    return result;
}
```

### Problem 3.25

```
long loop_while2(long a, long b) {
    long result = b;
    while (b > 0) {
        result = result * a;
        b = b - a;
    }
    return result;
}
```

### Problem 3.26

A. jump-to-middle translation was used for the loop

B.

```

long fun_a(unsigned long x) {
    long val = 0;
    while (x) {
        val ^= x;
        x >>= 1;
    }
    return val & 0x1;
}

```

C. The function computes the parity of the number of on bits in `x`, returning 1 if it's an odd number, returning 0 otherwise.

### Problem 3.27

```

long fact_for_while(long n) {
    long result = 1;
    if (n < 2)
        goto done;
    long i = 2;
loop:
    result *= i;
    i++;
    if (i <= n)
        goto loop;
done:
    return result;
}

```

### Problem 3.28

A.

```

long fun_b(unsigned long x) {
    long val = 0;
    long i;
    for (i = 64; i; i--) {
        val *= 2;
        val |= (x & 0x1);
        x >>= 1;
    }
    return val;
}

```

B. The compiler knows that the loop will always run exactly 64 times, so the initial test and jump can be omitted.

C. This function reverses the bit representation of `x`.

### Problem 3.29

A. We would also skip incrementing  $i$ , resulting in an infinite loop.

```
long sum = 0;
long i = 0;

while (i < 10) {
    if (i & 1)
        continue; /* causes infinite loop */
    sum += i;
    i++;
}
```

B. We can fix it like this:

```
long sum = 0;
long i = 0;

while (i < 10) {
    if (i & 1)
        goto end_loop;
    sum += i;
end_loop:
    i++;
}
```

### Problem 3.30

A. Case labels:  $-1, 0, 1, 2, 4, 5, 7$

B. 0 and 7 as well as 2 and 4 label the same cases.

### Problem 3.31

```
void switcher(long a, long b, long c, long *dest) {
    long val;
    switch(a) {
    case 5:
        c = b ^ 15;
        /* Fall through */
    case 0:
        val = c + 112;
        break;
    case 2:
    case 7:
        val = (b + c) << 2;
        break;
    case 4:
        val = a;
        break;
    default:
        val = b;
    }
    *dest = val;
}
```

### Problem 3.32

| Label | PC       | Instruction | %rdi | %rsi | %rax | %rsp          | *%rsp    | Description               |
|-------|----------|-------------|------|------|------|---------------|----------|---------------------------|
| M1    | 0x400560 | callq       | 10   | —    | —    | 0x7fffffff820 | -        | Call first(10)            |
| F1    | 0x400548 | lea         | 10   | —    | —    | 0x7fffffff818 | 0x400565 | Entry of first            |
| F2    | 0x40054c | sub         | 10   | 11   | —    | 0x7fffffff818 | 0x400565 | Prepare argument for last |
| F3    | 0x400550 | callq       | 9    | 11   | —    | 0x7fffffff818 | 0x400565 | Call last(9, 11)          |
| L1    | 0x400540 | mov         | 9    | 11   | —    | 0x7fffffff810 | 0x400555 | Entry of last             |
| L2    | 0x400543 | imul        | 9    | 11   | 9    | 0x7fffffff810 | 0x400555 | Calculate return value    |
| L3    | 0x400547 | retq        | 9    | 11   | 99   | 0x7fffffff810 | 0x400555 | Return 99 from last       |
| F4    | 0x400555 | repz retq   | 9    | 11   | 99   | 0x7fffffff818 | 0x400565 | Return 99 from first      |
| M2    | 0x400565 | mov         | 9    | 11   | 99   | 0x7fffffff820 | -        | Resume main               |

### Problem 3.33

The params are (in order):

| Param | Type   | Register |
|-------|--------|----------|
| a     | int    | %edi     |
| b     | short  | %si      |
| u     | long * | %rdx     |
| v     | char * | %rcx     |

It's also possible to swap `a` and `u` with `b` and `v`, respectively.

### Problem 3.34

- A. Local values `a0` - `a5` are stored in callee-saved registers.
- B. Local values `a6` and `a7` are stored on the stack.
- C. Not all local values are stored in callee-saved registers because there are only six of them (`%r12` - `%r15`, `%rbp` and `rbx`).

### Problem 3.35

- A. The value of the parameter `x` is stored in callee-saved register `%rbx`.
- B.

```
long rfun(unsigned long x) {
    if (x == 0)
        return 0;
    unsigned long nx = x >> 2;
    long rv = rfun(nx);
    return x + rv;
}
```

### Problem 3.36

| Array | Element size | Total size | Start address | Element $i$ |
|-------|--------------|------------|---------------|-------------|
| S     | 2            | 14         | $x_S$         | $x_S + 2i$  |
| T     | 8            | 24         | $x_T$         | $x_T + 8i$  |
| U     | 8            | 48         | $x_U$         | $x_U + 8i$  |
| V     | 4            | 32         | $x_V$         | $x_V + 4i$  |
| W     | 8            | 32         | $x_W$         | $x_W + 8i$  |

### Problem 3.37

| Expression             | Type    | Value             | Assembly code                            |
|------------------------|---------|-------------------|------------------------------------------|
| <code>S+1</code>       | short * | $x_S + 2$         | <code>leaq 2(%rdx), %rax</code>          |
| <code>S[3]</code>      | short   | $M[x_S + 6]$      | <code>movw 6(%rdx), %ax</code>           |
| <code>&amp;S[i]</code> | short * | $x_S + 2i$        | <code>leaq (%rdx,%rcx,2), %rax</code>    |
| <code>S[4*i+1]</code>  | short   | $M[x_S + 8i + 2]$ | <code>movw 2(%rdx,%rcx,8), %ax</code>    |
| <code>S+i-5</code>     | short * | $x_S + 2i - 10$   | <code>leaq -10(%rdx,%rcx,2), %rax</code> |

### Problem 3.38

```

long sum_element(long i, long j)
i in %rdi, j in %rsi
sum_element:
    leaq    0(,%rdi,8), %rdx      8*i
    subq    %rdi, %rdx           7*i
    addq    %rsi, %rdx           7*i+j
    leaq    (%rsi,%rsi,4), %rax   5*j
    addq    %rax, %rdi           i+5*j
    moveq   Q(,%rdi,8), %rax     Q[i+5*j]
    addq    P(,%rdx,9), %rax     Q[i+5*j] + P[7*i+j]

```

Based on the assembly code the values are  $M = 5$  and  $N = 7$ .

### Problem 3.39

The address of element  $i, j$  in a nested array is

$$\&A[i][j] = x_A + L(C \cdot i + j)$$

where  $L$  is the element size in bytes and  $C$  is the number of columns.

Applying this to our pointers:

- **Aptr:**  $\&A[i][0] = x_A + 4(16i + 0) = x_A + 64i$
- **Bptr:**  $\&B[0][k] = x_B + 4(16 \cdot 0 + k) = x_B + 4k$
- **Bend:**  $\&B[16][k] = x_B + 4(16 \cdot 16 + k) = x_B + 1024 + 4k$

### Problem 3.40

```
void fix_set_diag_opt(fix_matrix A, int val) {
    int *Aptr = &A[0][0];
    long i = 0;
    long iend = N(N+1);
    do {
        Aptr[i] = val;
        i += (N+1); // advance by one row and one col
    } while (i != iend);
}
```

### Problem 3.41

A.

| Field | Offset |
|-------|--------|
| p     | 0      |
| s.x   | 8      |
| s.y   | 12     |
| next  | 16     |

B. In total the structure requires 24 bytes.

C.

```
void sp_init(struct prob *sp) {
    sp->s.x = sp->s.y;
    sp->p = &(sp->s);
    sp->next = sp;
}
```

### Problem 3.42

A.

```
long fun(struct ELE *ptr) {
    long sum = 0;
    while (ptr) {
        sum += ptr->v;
        ptr = ptr->p;
    }
    return sum;
}
```

B. ELE implements a linked list and fun computes the sum of all elements.

### Problem 3.43

| <i>expr</i>        | <i>type</i> | Code                                                                     |
|--------------------|-------------|--------------------------------------------------------------------------|
| up->t1.u           | long        | <pre> movq (%rdi), %rax movq %rax, (%rsi) </pre>                         |
| up->t1.v           | short       | <pre> movw 8(%rdi), %ax movw %ax, (%rsi) </pre>                          |
| &up->t1.w          | char *      | <pre> leaq 10(%rdi), %rax movq %rax, (%rsi) </pre>                       |
| up->t2.a           | int *       | <pre> movq %rdi, (%rsi) </pre>                                           |
| up->t2.a[up->t1.u] | int         | <pre> movq (%rdi), %rax movl (%rdi,%rax,4) %eax movl %eax, (%rsi) </pre> |
| *up->t2.p          | char        | <pre> movq 8(%rdi), %rax movb (%rax), %al movb %al, (%rsi) </pre>        |

### Problem 3.44

A.

| i | c | j | d  | Size | Alignment |
|---|---|---|----|------|-----------|
| 0 | 4 | 8 | 12 | 16   | 4         |

B.

| i | c | d | j | Size | Alignment |
|---|---|---|---|------|-----------|
| 0 | 4 | 5 | 8 | 16   | 8         |

C.

| w | c | Size | Alignment |
|---|---|------|-----------|
| 0 | 6 | 10   | 2         |

D.

| w | c  | Size | Alignment |
|---|----|------|-----------|
| 0 | 16 | 40   | 8         |

E.

| a | t  | Size | Alignment |
|---|----|------|-----------|
| 0 | 24 | 40   | 8         |

### Problem 3.45

A. & B.

| a | b | c  | d  | e  | f  | g  | h  | Size |
|---|---|----|----|----|----|----|----|------|
| 0 | 8 | 16 | 24 | 28 | 32 | 40 | 48 | 56   |

C. We can rearrange the fields like this to minimize wasted space:

| a | b | d  | f  | e  | c  | g  | h  | Size |
|---|---|----|----|----|----|----|----|------|
| 0 | 8 | 10 | 11 | 12 | 16 | 24 | 32 | 40   |

### Problem 3.46

A.

| Value                   | Label                               |
|-------------------------|-------------------------------------|
| 00 00 00 00 00 40 00 76 | Return address                      |
| 01 23 45 67 89 AB CD EF | Value of callee-saved register %rbx |
|                         |                                     |
|                         | buf = %rsp                          |
|                         |                                     |

B.

| Value                   | Label                               |
|-------------------------|-------------------------------------|
| 00 00 00 00 00 40 00 34 | Return address                      |
| 33 32 31 30 39 38 37 36 | Value of callee-saved register %rbx |
| 35 34 33 32 31 30 39 38 |                                     |
| 37 36 35 34 33 32 31 30 | buf = %rsp                          |
|                         |                                     |

C. The program attempts to return to the address `0x400034`. The two low bytes were overwritten by `0x34` (last char from stdin) and `0x00` (terminating null character).

D. The register %rbx will be corrupted after returning, since its saved value on the stack was overwritten and it is callee-saved, so the callee is not allowed to change it.

E. `strlen` returns the length without the null terminator, so `malloc` allocates one too few bytes. Also, `malloc` can return `NULL`, which is not safe to pass into `strcpy`.

### Problem 3.47

A.  $0xd754 - 0xb754 = 0x2000 = 2 \cdot 16^3 = 2^{13}$ .

B. It would take about  $\frac{2^{13}}{2^7} = 2^6 = 64$  attempts.

### Problem 3.48

A.

Without protector:

| Name         | Position in stack frame |
|--------------|-------------------------|
| buf          | (%rsp)                  |
| v            | 24(%rsp)                |
| canary value | -                       |

With protector:

| Name         | Position in stack frame |
|--------------|-------------------------|
| buf          | 16(%rsp)                |
| v            | 8(%rsp)                 |
| canary value | 40(%rsp)                |

B. In the rearranged variant the canary value is immediately adjacent to `buf`, so `v` cannot be corrupted.

### Problem 3.49

A.  $s_2$  is calculated by subtracting  $8n + 22$  rounded down to the closest multiple of 16 from  $s_1$ . If  $n$  is odd it will result in  $s_1 - (8n + 8)$ , otherwise  $s_1 - (8n + 16)$ .

B. We take  $s_2 + 7$  and round down to the closest multiple of 8 (alignment). This effectively rounds up  $s_2$  to the closest multiple of 8.

C.

| $n$ | $s_1$ | $s_2$ | $p$  | $e_1$ | $e_2$ |
|-----|-------|-------|------|-------|-------|
| 5   | 2065  | 2017  | 2024 | 1     | 7     |
| 6   | 2064  | 2000  | 2000 | 16    | 0     |

D.  $s_2$  is aligned to multiples of 16, and  $p$  is aligned to multiples of 8.

### Problem 3.50

| Expression | Mapping |
|------------|---------|
| val1       | d       |
| val2       | i       |
| val3       | l       |
| val4       | f       |

### Problem 3.51

| $T_x$  | $T_y$  | Instruction(s)                                                                    |
|--------|--------|-----------------------------------------------------------------------------------|
| long   | double | <code>vcvtsi2sdq %rdi, %xmm0, %xmm0</code>                                        |
| double | int    | <code>vcvtt2sd2si %xmm0, %eax</code>                                              |
| double | float  | <code>vunpcklpd %xmm0, %xmm0, %xmm0</code><br><code>vcvtpd2ps %xmm0, %xmm0</code> |
| long   | float  | <code>vcvtsi2ssq %rdi, %xmm0, %xmm0</code>                                        |
| float  | long   | <code>vcvttss2siq %xmm0, %rax</code>                                              |

### Problem 3.52

A.

```
double g1(double a, long b, float c, int d);
```

| Argument | Register |
|----------|----------|
| a        | %xmm0    |
| b        | %rdi     |
| c        | %xmm1    |
| d        | %esi     |

B.

```
double g2(int a, double *b, float *c, long d);
```

| Argument | Register |
|----------|----------|
| a        | %edi     |
| b        | %rsi     |

| Argument | Register |
|----------|----------|
| c        | %rdx     |
| d        | %rcx     |

C.

```
double g3(double *a, double b, int c, float d);
```

| Argument | Register |
|----------|----------|
| a        | %rdi     |
| b        | %xmm0    |
| c        | %esi     |
| d        | %xmm1    |

D.

```
double g4(float a, int *b, float c, double d);
```

| Argument | Register |
|----------|----------|
| a        | %xmm0    |
| b        | %rdi     |
| c        | %xmm1    |
| d        | %xmm2    |

### Problem 3.53

| arg1_t | arg2_t | arg3_t | arg4_t |
|--------|--------|--------|--------|
| int    | float  | long   | double |
| int    | long   | float  | double |

### Problem 3.54

```
double funct2(double w, int x, float y, long z) {
    return y*x - w/z;
}
```

### Problem 3.55

Low-order 4 bytes:

```
00 00 00 00
```

High-order 4 bytes:

```
40 40 00 00
```

Complete number in binary:

```
01000000 01000000 00000000 00000000 00000000 00000000 00000000 00000000
seeeeeee eeeeffff ffffffff ffffffff ffffffff ffffffff ffffffff ffffffff
```

So the sign is positive.

The exponent is  $1028 - 1023 = 5$ .

The fraction is 1.0 (with implied leading 1).

The value is thus  $2^5 \times 1.0 = 32.0$ .

### Problem 3.56

A.

Constant: `0x7fffffffffffffff`

Binary: `01111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111`

This bitmask applied with `&` calculates the absolute value of the `double`.

B.

This sets the register to all zeroes since `a ^ a = 0`.

C.

Constant: `0x8000000000000000`

This bitmask applied with `^` negates the value of the `double`.

### Problem 3.57

```
double funct3(int *ap, double b, long c, float *dp) {
    int a = *ap;
    float d = *dp;
    if (a < b) {
        return c*d;
    } else {
        return c + 2*d;
    }
}
```

### Problem 3.58

```
long decode2(long x, long y, long z) {
    y -= z;
    x *= y;
    return ((y << 63) >> 63) ^ x;
}
```

### Problem 3.59

```
dest in %rdi, x in %rsi, y in %rdx
store_prod:
    movq    %rdx, %rax          y = yl
    cqto                                %rdx:%rax = yh:yl
    movq    %rsi, %rcx
    sarq    $63, %rcx           xh = sign of x
    imulq   %rax, %rcx          r = yl * xh
    imulq   %rsi, %rdx          s = yh * xl
    addq    %rdx, %rcx          r + s
    mulq    %rsi                %rdx:%rax = xl * yl = t
    addq    %rcx, %rdx          r + s + th
    movq    %rax, (%rdi)        store pl in memory
    movq    %rdx, 8(%rdi)       store ph in memory
    ret
```

`r` and `s` are corrections for signed multiplication which we have to consider because `mulq` does unsigned multiplication (and only the the lower 64 bits (= size of factors) are equivalent between two's complement and unsigned).

### Problem 3.60

A.

| Value  | Register |
|--------|----------|
| x      | %rdi     |
| n      | %esi     |
| result | %rax     |
| mask   | %rdx     |

B. The initial values of `result` and `mask` are 0 and 1, respectively.

C. The test condition is `mask != 0`.

D. `mask` is arithmetically shifted left by `n`.

E. The result is OR'ed with `x & mask` .

F.

```
long loop(long x, int n) {  
    long result = 0;  
    long mask;  
    for (mask = 1; mask != 0; mask <= n) {  
        result |= (x & mask);  
    }  
    return result;  
}
```

### Problem 3.61

```
long cread(long *xp) {  
    long zero = 0;  
    if (!xp) xp = &zero;  
    return *xp;  
}
```

### Problem 3.62

```
typedef enum {MODE_A, MODE_B, MODE_C, MODE_D, MODE_E} mode_t;

long switch3(long *p1, long *p2, mode_t action) {
    long result = 0;
    switch(action) {
        case MODE_A:
            result = *p2;
            *p2 = *p1;
            break;
        case MODE_B:
            *p1 += *p2;
            result = *p1;
            break;
        case MODE_C:
            *p1 = 59;
            result = *p2;
            break;
        case MODE_D:
            *p1 = *p2;
            /* fall through */
        case MODE_E:
            result = 27;
            break;
        default:
            result = 12;
    }
    return result;
}
```

### Problem 3.63

```

long switch_prob(long x, long n) {
    long result = x;
    switch(n) {
        case 60:
        case 62:
            result <= 3;
            break;
        case 63:
            result >= 3;
            break;
        case 64:
            result *= 15;
            /* fall through */
        case 65:
            result *= result;
            /* fall through */
        case 61:
            result += 75;
    }
    return result;
}

```

### Problem 3.64

A. Formula for location of array element `A[i][j][k]` :

$$\&D[i][j][k] = x_D + 8(iST + jT + k)$$

where  $R$ ,  $S$  and  $T$  are the sizes of the nested arrays: `long D[R][S][T]` .

B.

```

long store_ele(long i, long j, long k, long *dest)
i in %rdi, j in %rsi, k in %rdx, dest in %rcx
store_ele:
    leaq    (%rsi,%rsi,2), %rax    3*j
    leaq    (%rsi,%rax,4), %rax    4(3*j)+j = 13*j
    movq    %rdi, %rsi            j = i
    salq    $6, %rsi              64*i
    addq    %rsi, %rdi             65*i
    addq    %rax, %rdi             13*j + 65*i
    addq    %rdi, %rdx             13*j + 65*i + k
    movq    A(,%rdx,8), %rax       read array element
    movq    %rax, (%rcx)           store at dest
    movl    $3640, %eax           return sizeof(A)
    ret

```

The index is calculated as  $5 \cdot 13 \cdot i + 13 \cdot j + k$  which suggests that  $S = 5$  and  $T = 13$ .

We also know that  $8RST = 3640$ , so  $R = \frac{3640}{8 \cdot 5 \cdot 13} = 7$ .

### Problem 3.65

- A. The pointer to element `A[i][j]` is in `%rdx`.
- B. The pointer to element `A[j][i]` is in `%rax`.
- C. `%rax` is incremented by  $120 = 15 \cdot 8$  every iteration, so each row must have 15 elements, thus  $M = 15$ .

### Problem 3.66

The address that we read each iteration ( `%rcx` ) is incremented by  $8(4n + 1)$  so `NC` must be defined as  $4n + 1$ .

The loop condition compares `i` (in register `%rdx`) with `NR(n)` (in register `%rdi`). Looking at `%rdi` we know that `NR(n)` must be  $3n$ .

### Problem 3.67

A.

| Offset from <code>%rsp</code> | Description         |
|-------------------------------|---------------------|
| 96                            |                     |
| 88                            |                     |
| 80                            | <code>r.q</code>    |
| 72                            | <code>r.u[1]</code> |
| 64                            | <code>r.u[0]</code> |
| 56                            |                     |
| 48                            |                     |
| 40                            |                     |
| 32                            |                     |
| 24                            | <code>z</code>      |
| 16                            | <code>s.p</code>    |
| 8                             | <code>s.a[1]</code> |
| 0                             | <code>s.a[0]</code> |

- B. `eval` passes a pointer to the space on the stack allocated for `r` to `process`.

C. Elements of `s` are accessed by calculating the address offsets for each element. It starts at `%rsp+8` because the return address is stored at `%rsp`.

D. Values on the return structure are set based on the address passed in as `%rdi`.

E. (see A.)

F. Structure function arguments are passed as pointers on the stack. Space for a function return value that is a structure is allocated by the caller and a pointer to it is passed as a hidden argument.

### Problem 3.68

We see that to access `t` an offset of 8 is used. `t` is of type `int` so it must be aligned to an offset divisible by 4. This means  $B$  must be between 5 and 8.

Similarly, to access `u` an offset of 32 is used. `s` starts at offset 12. So `s` and possible padding take up 20 bytes. `u` is of type `long` which means it must be aligned to an offset divisible by 8. `s` has type `short` so each element takes up 2 bytes. From this we conclude that  $A$  must be between 7 and 10.

From the offset used to access `y` we see that `x` plus possible padding takes up 184 bytes. `y` has to be aligned to an offset divisible by 8. `x` itself is of type `int` so each element uses 4 bytes. This means the array itself must be either 184 bytes long or 180 bytes plus 4 bytes of padding. Which means it has either 45 or 46 elements. The only possible way to get either of these by multiplying possible values for  $A$  and  $B$  is  $9 \cdot 5 = 45$ .

So we can conclude that  $A = 9$  and  $B = 5$ .

### Problem 3.69

```
i in %rdi, bp in %rsi
test:
    mov     0x120(%rsi),%ecx      read last (offset 288)
    add     (%rsi),%ecx          first + last
    lea     (%rdi,%rdi,4),%rax    5*i
    lea     (%rsi,%rax,8),%rax    8*5*i
    mov     0x8(%rax),%rdx       read idx
    movslq  %ecx,%rcx           extend to 64 bits
    mov     %rcx,0x10(%rax,%rdx,8)
    retq
```

A. The size of `a_struct` is 40 bytes. Using the offset of `last` we have  $CNT = \frac{288-8}{40} = 7$ .

B.

```
typedef struct {
    long idx;
    long x[4];
} a_struct;
```

### Problem 3.70

A.

| Field   | Offset |
|---------|--------|
| e1.p    | 0      |
| e1.y    | 8      |
| e2.x    | 0      |
| e2.next | 8      |

B. In total the structure requires 16 bytes.

C.

```
up in %rdi
proc:
    movq    8(%rdi), %rax    read e2.next (or e1.y)
    movq    (%rax), %rdx     *e2.next
    movq    (%rdx), %rdx     *(e2.next)->p
    subq    8(%rax), %rdx    subtract e2.next->e1.y
    movq    %rdx, (%rdi)    store in e2.x
    ret
```

```
union ele {
    struct {
        long *p;
        long y;
    } e1;
    struct {
        long x;
        union ele *next;
    } e2;
};

void proc (union ele *up) {
    up->e2.x = *(up->e2.next->e1.p) - up->e2.next->e1.y;
}
```

### Problem 3.71

```
#include <stdio.h>
#include <string.h>

#define BUFSIZE 64

void good_echo() {
    char buf[BUFSIZE];
    while (fgets(buf, BUFSIZE, stdin)) {
        if (fputs(buf, stdout) == EOF)
            return; // error while writing

        int i;
        for (i = 0; buf[i] && buf[i] != '\n'; i++)
            ;
        if (buf[i] == '\n')
            return;

        // or using standard library
        // if (strchr(buf, '\n'))
        //     return;
    }
}
```

### Problem 3.72

```
long vframe(long n, long idx, long *q) {
    long i;
    long *p[n];
    p[0] = &i;
    for (i = 1; i < n; i++) {
        p[i] = q;
    }
    return *p[idx];
}
```

- A. We calculate  $8n + 30$  and then round down to the closest multiple of 16. For even  $n$  we get  $8n + 16$  and for odd  $n$  we get  $8n + 24$ .  $s_2$  is calculated by subtracting this amount from  $s_1$ .
- B.  $p$  is computed by rounding  $s_2$  up to the closest multiple of 16 (by adding 15 and then rounding down).
- C. *skipped*
- D. *skipped*

### Problem 3.73

```
.globl find_range
find_range:
    vxorps %xmm1, %xmm1, %xmm1 # set %xmm1 = 0
    vucomiss %xmm1, %xmm0
    jp .OTHER
    jb .NEG
    ja .POS
    movq $1, %rax
    ret
.NEG:
    movq $0, %rax
    ret
.POS:
    movq $2, %rax
    ret
.OTHER:
    movq $3, %rax
    ret
```

Test all possible inputs by calling from C:

```
#include <stdint.h>
#include <stdio.h>

typedef enum { NEG, ZERO, POS, OTHER } range_t;

extern range_t find_range(float x);

long r[4];

int main() {
    uint32_t bits = 0;
    while (1) {
        float f = *((float *)&bits); // preserve bits instead of value
        r[find_range(f)]++;
        if (bits == UINT32_MAX)
            break;
        bits++;
    }

    printf("NEG : %ld\n", r[0]);
    printf("ZERO : %ld\n", r[1]);
    printf("POS : %ld\n", r[2]);
    printf("OTHER: %ld\n", r[3]);
    printf("TOTAL: %ld\n", r[0] + r[1] + r[2] + r[3]);
}
```

### Problem 3.74

```
.globl find_range
find_range:
    vxorps %xmm1, %xmm1, %xmm1 # set %xmm1 = 0
    vucomiss %xmm1, %xmm0
    movq $1, %rax
    movq $0, %rdi
    cmovb %rdi, %rax
    movq $2, %rdi
    cmova %rdi, %rax
    movq $3, %rdi
    cmovp %rdi, %rax
    ret
```

(use C code above for testing)

### Problem 3.75

A. Complex arguments are passed in two registers, one containing the real and one containing the imaginary part.

B. Similarly for return values: real part in `%xmm0` , imaginary part in `%xmm1` .

## Chapter 4 Processor Architecture

### Problem 4.1

| Instruction                       | Encoding                      |
|-----------------------------------|-------------------------------|
| <code>irmovq \$15,%rbx</code>     | 30 f3 0f 00 00 00 00 00 00 00 |
| <code>rrmovq %rbx,%rcx</code>     | 20 31                         |
| <code>rmmovq %rcx,-3(%rbx)</code> | 40 13 fd ff ff ff ff ff ff ff |
| <code>addq %rbx,%rcx</code>       | 60 31                         |
| <code>jmp loop</code>             | 70 0c 01 00 00 00 00 00 00 00 |

### Problem 4.2

| Byte Sequence                 | Instruction Sequence                |
|-------------------------------|-------------------------------------|
| A.                            |                                     |
| 0x100: 30f3fcffffffffffffffff | <code>irmovq \$-4,%rbx</code>       |
| 0x10a: 4063000800000000000000 | <code>rmmovq %rsi,2048(%rbx)</code> |

| Byte Sequence               | Instruction Sequence                                                          |
|-----------------------------|-------------------------------------------------------------------------------|
| 0x114: 00                   | halt                                                                          |
| <b>B.</b>                   |                                                                               |
| 0x200: a06f                 | pushq %rsi                                                                    |
| 0x202: 800c0200000000000000 | call proc                                                                     |
| 0x20b: 00                   | halt                                                                          |
| 0x20c:                      | proc:                                                                         |
| 0x20c: 30f30a00000000000000 | irmovq \$10,%rbx                                                              |
| 0x216: 90                   | ret                                                                           |
| <b>C.</b>                   |                                                                               |
| 0x300: 50540700000000000000 | mrmovq 7(%rsp),%rbp                                                           |
| 0x30a: 10                   | nop                                                                           |
| 0x30b: f0                   | invalid instruction                                                           |
| 0x30c: b01f                 | popq %rcx                                                                     |
| <b>D.</b>                   |                                                                               |
| 0x400:                      | loop:                                                                         |
| 0x400: 6113                 | subq %rcx,%rbx                                                                |
| 0x402: 73000400000000000000 | je loop                                                                       |
| 0x40c: 00                   | halt                                                                          |
| <b>E.</b>                   |                                                                               |
| 0x500: 6362                 | xorq %rsi,%rdx                                                                |
| 0x502: a0f0                 | f means “no register” but pushq requires one (but the second part must be f ) |

### Problem 4.3

```
long sum(long *start, long count)
start in %rdi, count in %rsi
sum:
    xorq %rax,%rax        sum = 0
    andq %rsi,%rsi        Set CC
    jmp test              Goto test
loop:
    mrmovq (%rdi),%r10     Get *start
    addq %r10,%rax         Add to sum
    iaddq $8,%rdi          start++
    iaddq $-1,%rsi         count-- and set CC
test:
    jne loop              Stop when 0
    ret
```

### Problem 4.4

```
long rsum(long *start, long count)
start in %rdi, count in %rsi
rsum:
    irmovq $0,%rax        sum = 0
    andq %rsi,%rsi        Set CC
    jg rec                Stop if count <= 0
    ret
rec:
    pushq %rbx            Save callee-saved register
    mrmovq (%rdi),%rbx    Read *start
    irmovq $1,%r8
    subq %r8,%rsi         count--
    irmovq $8,%r8
    addq %r8,%rdi         start++
    call rsum             Recurse
    addq %rbx,%rax        sum += *start
    popq %rbx             Restore callee-saved register
    ret
```

## Problem 4.5

```
long absSum(long *start, long count)
start in %rdi, count in %rsi
absSum:
    irmovq $8,%r8          Constant 8
    irmovq $1,%r9          Constant 1
    xorq %rax,%rax         sum = 0
    andq %rsi,%rsi         Set CC
    jmp test               Goto test
loop:
    mrmovq (%rdi),%r10      Get *start
    andq %r10,%r10         Set CC
    jl sub
add:
    addq %r10,%rax         Add positive number to sum
    jmp cont
sub:
    subq %r10,%rax         Subtract negative number from sum
cont:
    addq %r8,%rdi          start++
    subq %r9,%rsi          count-- and set CC
test:
    jne loop               Stop when 0
    ret
```

## Problem 4.6

```
long absSum(long *start, long count)
start in %rdi, count in %rsi
absSum:
    irmovq $8,%r8          Constant 8
    irmovq $1,%r9          Constant 1
    xorq %rax,%rax         sum = 0
    andq %rsi,%rsi         Set CC
    jmp test               Goto test
loop:
    mrmovq (%rdi),%r10      x = *start
    xorq %r11,%r11         Set 0
    subq %r10,%r11         -x
    cmovg %r11,%r10        if -x > 0 then x = -x
    addq %r10,%rax         Add to sum
    addq %r8,%rdi          start++
    subq %r9,%rsi          count-- and set CC
test:
    jne loop               Stop when 0
    ret
```

### Problem 4.7

It implies that the register value is read before decrementing the stack pointer.

### Problem 4.8

It implies that the value read from memory is written to `%rsp` instead of incrementing it. It is equivalent to:

```
mrmovq (%rsp),%rsp
```

### Problem 4.9

We can define `xor` like this:

```
bool xor = (a && !b) || (!a && b)
```

It relates to `eq` like this:

```
bool xor = !eq
```

### Problem 4.10

We can implement word-level equality using `xor` very similarly to the `==` implementation. We combine 64 `xor` circuits comparing each individual bit of `a` and `b`, combine all the outputs with `or`, and then negate the result.

### Problem 4.11

```
word Min3 = [  
  A <= B && A <= C : A;  
  A <= C           : B;  
  1                : C;  
]
```

Using `B <= C` as the second condition also works.

### Problem 4.12

Circuit for computing the median of three numbers:

```

word Med3 = [
  B <= A && A <= C : A;
  C <= A && A <= B : A;
  A <= B && B <= C : B;
  C <= B && B <= A : B;
  1 : C;
]

```

### Problem 4.13

| Stage      | Generic                                                                                                                                                                                 | Specific                                                                                                                                                                                      |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <code>irmovq V, rB</code>                                                                                                                                                               | <code>irmovq \$128, %rsp</code>                                                                                                                                                               |
| Fetch      | $\text{icode:ifun} \leftarrow M_1[\text{PC}]$<br>$\text{rA:rB} \leftarrow M_1[\text{PC} + 1]$<br>$\text{valC} \leftarrow M_8[\text{PC} + 2]$<br>$\text{valP} \leftarrow \text{PC} + 10$ | $\text{icode:ifun} \leftarrow M_1[0x016] = 3 : 0$<br>$\text{rA:rB} \leftarrow M_1[0x017] = f : 4$<br>$\text{valC} \leftarrow M_8[0x018] = 128$<br>$\text{valP} \leftarrow 0x016 + 10 = 0x020$ |
| Decode     | -                                                                                                                                                                                       | -                                                                                                                                                                                             |
| Execute    | $\text{valE} \leftarrow 0 + \text{valC}$                                                                                                                                                | $\text{valE} \leftarrow 0 + 128 = 128$                                                                                                                                                        |
| Memory     | -                                                                                                                                                                                       | -                                                                                                                                                                                             |
| Write back | $R[\text{rB}] \leftarrow \text{valE}$                                                                                                                                                   | $R[\%rsp] \leftarrow 128$                                                                                                                                                                     |
| PC update  | $\text{PC} \leftarrow \text{valP}$                                                                                                                                                      | $\text{PC} \leftarrow 0x020$                                                                                                                                                                  |

The instruction sets `%rsp` to 128 and increments the program counter by 10.

### Problem 4.14

| Stage      | Generic                                                                                                                                 | Specific                                                                                                                            |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|            | <code>popq rA</code>                                                                                                                    | <code>popq %rax</code>                                                                                                              |
| Fetch      | $\text{icode:ifun} \leftarrow M_1[\text{PC}]$<br>$\text{rA:rB} \leftarrow M_1[\text{PC} + 1]$<br>$\text{valP} \leftarrow \text{PC} + 2$ | $\text{icode:ifun} \leftarrow M_1[0x02c] = b : 0$<br>$\text{rA:rB} \leftarrow M_1[0x02d] = 0 : f$<br>$\text{valP} \leftarrow 0x02e$ |
| Decode     | $\text{valA} \leftarrow R[\%rsp]$<br>$\text{valB} \leftarrow R[\%rsp]$                                                                  | $\text{valA} \leftarrow R[\%rsp] = 120$<br>$\text{valB} \leftarrow R[\%rsp] = 120$                                                  |
| Execute    | $\text{valE} \leftarrow \text{valB} + 8$                                                                                                | $\text{valE} \leftarrow 120 + 8 = 128$                                                                                              |
| Memory     | $\text{valM} \leftarrow M_8[\text{valA}]$                                                                                               | $\text{valM} \leftarrow M_8[120] = 9$                                                                                               |
| Write back | $R[\%rsp] \leftarrow \text{valE}$<br>$R[\text{rA}] \leftarrow \text{valM}$                                                              | $R[\%rsp] \leftarrow 128$<br>$R[\%rax] \leftarrow 9$                                                                                |
| PC update  | $\text{PC} \leftarrow \text{valP}$                                                                                                      | $\text{PC} \leftarrow 0x02e$                                                                                                        |

The instruction sets `%rax` to 9, `%rsp` to 128 and increments the program counter by 2.

### Problem 4.15

Effect of `pushq %rsp` :

- push the old stack pointer to the stack (using the new (decremented) pointer as address)
- decrement stack pointer by 8
- increment the PC by 2

This conforms to the desired behavior of Y86-64, since it pushes the old value to the stack.

### Problem 4.16

`popq %rsp` has the effect of storing the value from the stack in `%rsp` (not the decremented stack pointer). This is the desired behavior for Y86-64.

### Problem 4.17

| Stage      | <code>cmovXX rA, rB</code>                                                                                                              |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Fetch      | $\text{icode:ifun} \leftarrow M_1[\text{PC}]$<br>$\text{rA:rB} \leftarrow M_1[\text{PC} + 1]$<br>$\text{valP} \leftarrow \text{PC} + 2$ |
| Decode     | $\text{valA} \leftarrow R[\text{rA}]$                                                                                                   |
| Execute    | $\text{valE} \leftarrow 0 + \text{valA}$<br>$\text{Cnd} \leftarrow \text{Cond}(\text{CC}, \text{ifun})$                                 |
| Memory     | -                                                                                                                                       |
| Write back | $\text{if}(\text{Cnd}) R[\text{rB}] \leftarrow \text{valE}$                                                                             |
| PC update  | $\text{PC} \leftarrow \text{valP}$                                                                                                      |

### Problem 4.18

| Stage      | Generic                                                                                                                                | Specific                                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|            | <code>call Dest</code>                                                                                                                 | <code>call 0x041</code>                                                                                                          |
| Fetch      | $\text{icode:ifun} \leftarrow M_1[\text{PC}]$<br>$\text{valC} \leftarrow M_8[\text{PC} + 1]$<br>$\text{valP} \leftarrow \text{PC} + 9$ | $\text{icode:ifun} \leftarrow M_1[0x037] = 8:0$<br>$\text{valC} \leftarrow M_8[0x038] = 0x041$<br>$\text{valP} \leftarrow 0x040$ |
| Decode     | $\text{valB} \leftarrow R[\%rsp]$                                                                                                      | $\text{valB} \leftarrow R[\%rsp] = 128$                                                                                          |
| Execute    | $\text{valE} \leftarrow \text{valB} + (-8)$                                                                                            | $\text{valE} \leftarrow 128 + (-8) = 120$                                                                                        |
| Memory     | $M_8[\text{valE}] \leftarrow \text{valP}$                                                                                              | $M_8[120] \leftarrow 0x040$                                                                                                      |
| Write back | $R[\%rsp] \leftarrow \text{valE}$                                                                                                      | $R[\%rsp] \leftarrow 120$                                                                                                        |
| PC update  | $\text{PC} \leftarrow \text{valC}$                                                                                                     | $\text{PC} \leftarrow 0x041$                                                                                                     |

This instruction decreases the stack pointer by 8, sets the PC to the destination address of the procedure to call and pushes the return address to the stack.

### Problem 4.19

```
bool need_calC = icode in { IIRMOVQ, IRMMOVQ, IMRMVQ, IJXX, ICALL };
```